

A Multimodal Decision–Fusion Network Approach for Activity Recognition in Firefighter Self-Contained Breathing Apparatus Endurance Training

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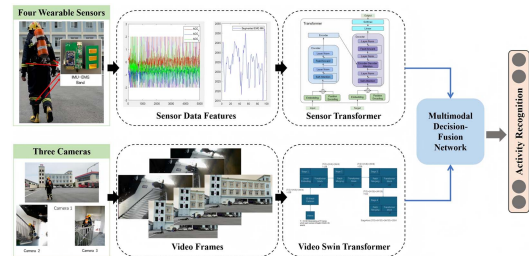
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Abstract—Insufficient training in firefighting techniques increases the risk of injuries and fatalities among firefighters. Human activity recognition methods show promising potential for performance monitoring and evaluation. However, the present studies focus mainly on individual modalities. This limited scope presents challenges in effectively distinguishing between intricate tasks, such as those encountered in firefighting operations. This study introduces an innovative multimodal decision–fusion network designed to overcome this limitation. This is achieved by integrating vision data sourced from three distinct cameras and sensor data collected from four wearable devices. The proposed network combines a vision-focused Video Swin network with a sensor-driven Sensor Transformer network where the results show that the use of only vision-based methods is insufficient to accurately classify firefighting training activities. The proposed decision–fusion network improves classification with a mean F1-score of 95.73%, outperformed the existing hybrid machine learning network.



Index Terms—Sensor applications, activity recognition, deep learning (DL), multimodal fusion, sensor fusion, transformer, video processing.

I. INTRODUCTION

The firefighting profession is highly demanding and hazardous, requiring substantial physical and mental resilience along with specialized skills for effective rescue operations. China's annual firefighting report [1] documented 111 firefighter casualties between 2010 and 2020, with an average age of 27 years. Insufficient training and experience contribute to injuries by leading to poor decision making [2]. Improving rescue techniques, physical endurance, and tactical proficiency requires specialized training. The scientific detail of the training methodologies significantly affects the physical capabilities and operational effectiveness of the firefighters. Studies show that many fire brigades lack innovative firefighting training, often relying on conventional military-style approaches that can reduce firefighter motivation [3]. Furthermore, traditional training methods are associated with increased firefighter fatigue [4].

Human activity recognition (HAR) improves firefighter training through various means. This method provides trainers with instant feedback to monitor firefighter actions during training, improve safety, and evaluate performance. Integrating HAR with the Internet of Things also enables remote supervision and assessment by trainers. A data-driven HAR approach records firefighter performance data over time, facilitating personalized training based on analysis of past sessions.

Recent studies on HAR are classified mainly as vision-based and sensor-based approaches [5]. Deep learning (DL) methods have been shown to be effective in vision-based HAR. For example, Carreira and Zisserman [6] proposed an inflated 3-D convolutional neural network (ConvNet), demonstrating improved performance over conventional 3-D convolutional neural network (CNN) on several multimotion recognition benchmarks. Liu et al. [7] proposed a Transformer-based architecture that captures temporal and spatial features through its layer architecture. In addition, Liu et al. [8] proposed a Swin transformer-adapted architecture for video recognition that utilized spatiotemporal locality to achieve a better speed-accuracy tradeoff and state-of-the-art performance on various video recognition benchmarks. This network provides high accuracy in visual image classification and reduces the resource consumption of the graphics processing unit. However, vision-based approaches are constrained in practical firefighting use due to the difficulty in continuously capturing firefighter motion and the requirement for an unobstructed view.

Instead, many researchers focus on sensor-based approaches that utilize sensor data worn on specific body locations. The inertial measurement unit (IMU) is the most widely used sensor in HAR, measuring acceleration and angular velocity at the body location. Many studies have focused on combining DL networks with multimodal network structures, particularly CNN with long short-term memory network [9]. Studies, such as [10] and [11], have addressed the high computational cost of DL networks by proposing lightweight CNNs. A hybrid machine learning (HML) approach [12] has been proposed for sensor-based HAR through fusion of multiple classifiers for HAR

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TABLE 1. Overview of SCBA Endurance Training Activities

Activity	Description
A1	Use a hammer to hit a tire.
A2	Drag dummy.
A3	Move the nozzle with a water-supplied hose.
A4	Climb four floors while carrying two trays of hoses.
A5	Drag the hose via the window.
A6	Descend from the fourth floor with two trays of hoses.
A7	Perform usual walking movement.
A8	Stationary actions.

classification. The results show that data fusion of multiple IMUs and surface electromyography (sEMG) significantly improves classification performance, achieving a recall of 91.32% that outperforms recent work. However, the use of wearable sensors faces mechanical impedance and noise that limit classification accuracy and efficiency in practical applications.

Therefore, based on our existing work [12], this study aims to further improve the classifier to distinguish firefighting activities. Unlike existing methods that rely on a single modality, this study proposes a novel multimodal fusion approach utilizing decision-level fusion between a Video Swin Transformer for video analysis and a Sensor Transformer network for sensor features. Vision-based approaches are limited by narrow fields of view and occlusion in dynamic firefighting scenarios, while sensor-based methods have difficulty in distinguishing fine operations; integrating both modalities addresses these individual limitations and significantly improves recognition performance for robust practical applications.

II. METHODOLOGY

A. Experiment Design

This study uses the SFTAA dataset collected in [12]. The dataset consists of self-contained breathing apparatus (SCBA) endurance training performed by 18 firefighters from the Huzhou Fire Brigade in Zhejiang province, China. SCBA endurance training consisted of six specialized activities (A1–A6), one regular movement (A7) and one stationary activity (A8), as shown in Table 1. Each firefighter completed the entire SCBA endurance training consecutively, comprising activities A1–A6, starting with a full oxygen capacity of 28 mPa. Initially, firefighters completed a training cycle that included activities A1–A6, consecutively. Firefighters then continued with further cycles as long as their SCBA oxygen level remained above 8 mPa (without triggering a warning). The SCBA alert triggered indicated completion of the training, and the firefighter moved to activity A8 where the firefighter rested for 2 and 3 min and finally removed their gear. Based on the training evaluation standard, each firefighter was required to complete at least 20 activities, equating to a minimum of three cycles. Generally, the duration of each experiment varied according to individual oxygen consumption rates, with a range of 15–25 min.

Fig. 1 illustrates the overview of the experimental setup for SCBA endurance training. Four IMU-electromyography modules were worn on the left and right forearms and calves, measuring 6-DoF motion data and sEMG signals at 100 Hz. Window segmentation was applied with a 2-s window length and a 1.5-s overlap. In total, 31 features were extracted, comprising 25 time-domain and six frequency-domain features. In addition, three CCTV cameras were installed on the empty field (camera 1), the first floor (camera 2), and the fourth floor (camera 3) of the training tower, allowing capture of all SCBA endurance training activities. Since data augmentation is effective in improving



Fig. 1. Experimental setup of the SCBA endurance training captured by camera 1.

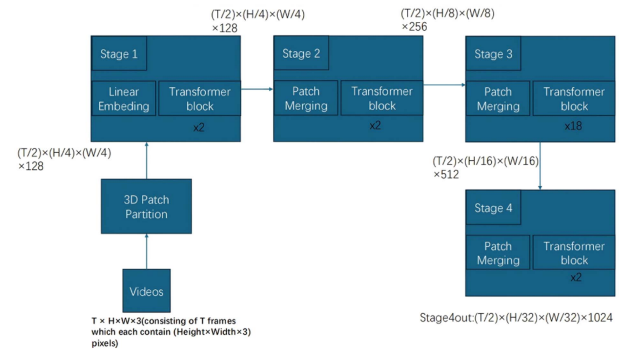


Fig. 2. Block diagram illustrating the architecture of the Video Swin Transformer network.

classifier robustness [13], this study applied flipping to video data. The videos were split into training and testing sets, and their file paths along with assigned class labels. The random shuffling of these data frames ensured that diverse batches were gathered during training to prevent the network from encountering sequential patterns that could introduce bias in its learning process.

The study followed ethical guidelines, obtained approval from the university's research ethics committee and complied with the university's code of conduct.

B. Video Swin Transformer Network

A pretrained Video Swin network, configured as in [8], served as the backbone feature extractor. This architecture extends the Swin Transformer into the spatio-temporal domain, capturing spatial and temporal dependencies through hierarchical patch embeddings and shifted window operations. Fig. 2 shows that each of the input videos (with T frames and resolution $H \times W$) were divided into 3-D patches without overlap. These patches were linearly embedded into a 128-channel feature space. They were then processed through multiple stages, each featuring Transformer blocks and patch merging operations. This process increases the feature dimensionality while reducing the spatial and temporal resolution.

Specifically, stage 1 outputs a feature size of $(T/2) \times (H/4) \times (W/4)$ with 128 channels, and the process was repeated in stages 2 and 3. In stage 4, the output is compressed in space and time dimensions, but enriched in semantics, with dimensions $(T/2) \times (H/32) \times (W/32)$ and 1024 channels. This hierarchical method preserves local spatio-temporal patterns and integrates them into global action representations. The backbone is initialized with weights pretrained on the

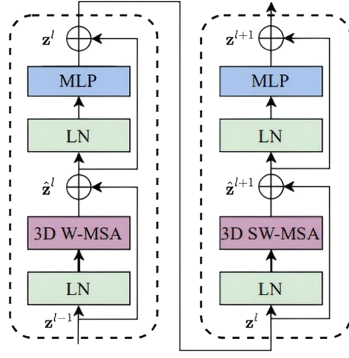


Fig. 3. Illustration of two sequential blocks in the Video Swin Transformer network.

“something-something” video dataset [14], which provides spatio-temporal features. This transfer learning strategy accelerates convergence and improves performance in the SCBA endurance training dataset. To maintain stability, the backbone layers are frozen during training in the SFTAA dataset.

A lightweight classification head, comprising a global average pooling (GAP) layer and a dense layer, was added to the backbone output. The GAP layer aggregates spatial and temporal information into a fixed-length feature vector, summarizing the activity features extracted by the Video Swin backbone. The dense layer maps these features to the target classes (eight SCBA endurance training activities). The classification head was newly trained to focus on the present domain, maintaining the stability of the backbone.

The Video Swin Transformer blocks are central to this architecture. They combine multihead self-attention (MSA), normalization layers, and feedforward multilayer perceptrons (MLPs). These blocks use 3-D window-based MSA and shifted window MSA to capture short-range and long-range dependencies. Fig. 3 illustrates two successive Video Swin Transformer blocks. The shifting mechanism allows the network to integrate global information across layers.

C. Sensor Transformer Network

A lightweight Transformer-based architecture classifies SCBA endurance training activities based on sensor data. The proposed network consists mainly of an input layer, an MSA block, a feedforward block, and a classification head, following the encoder structure of the Transformer network [15]. The respective functions are as follows.

- 1) *Input layer*: Receives the preprocessed sensor data, structured as a single time step with multiple sensor features.
- 2) *MSA block*: Applies self-attention to the feature dimension. Parallel execution of multiple attention heads effectively captures long-range dependencies. The attention output is combined with the original input through a residual connection and layer normalization to stabilize the training.
- 3) *Feedforward block*: Composed of two linear transformations with a rectified linear unit (ReLU) activation. A residual connection and layer normalization facilitate gradient flow.
- 4) *Classification head*: Extracts the time-step representation and passes it through a dense layer with softmax activation to produce the probability distribution over the target classes.

D. Multimodal Decision-Fusion Network

The different behavioral patterns and movement constraints in firefighting, such as heavy equipment, present unique challenges absent

TABLE 2. Performance Evaluation With Different Values of α

α	0.1	0.3	0.5	0.6	0.7	0.8	0.9
acc (%)	92.25	92.88	94.21	94.33	94.57	94.09	93.53

TABLE 3. Comparative Performance Analysis of HAR Networks

Methods	Precision	Recall	F1-score
HML [12]	90.94	93.48	92.19
Video Swin	88.41	87.92	88.10
Sensor Transformer	91.24	94.84	93.31
Decision-Fusion	96.67	96.13	95.73

in the recognition of daily activities, resulting in the lack of dedicated benchmark networks. The decision-fusion network effectively addresses these challenges by providing robustness to partial failure [16] and allows efficient adaptation by retraining the relevant subclassifier when the sensors are modified. The network components are detailed as follows.

- 1) *Sensor Transformer network*: A lightweight Transformer trained on sensor data. The input is shaped to match the expected sequence format of $(1, F)$, where F represents the number of features.
- 2) *Video Swin Transformer network*: A Swin Transformer backbone was altered to employ pretrained weights and a classification head, with the configuration as follows.
 - a) Input shape of $(T, H, W, 3)$, where $T = 32$ frames and $H = W = 224$.
 - b) Window size of $(16, 7, 7)$.
 - c) GAP followed by a fully connected layer.
 - d) Softmax output with eight classes.

A weighted decision-fusion strategy integrates the strengths of both networks. The final prediction probabilities are determined as follows:

$$P_{\text{final}} = \alpha \cdot P_{\text{sensor}} + (1 - \alpha) \cdot P_{\text{video}} \quad (1)$$

where P_{sensor} denotes the softmax output of the sensor-based Transformer network, P_{video} denotes the softmax output of the vision-based Video Swin Transformer network, and $\alpha \in [0, 1]$ is the fusion weight controlling the contribution of each modality. The final predicted class is determined by the index of the maximum value in P_{final} , as follows:

$$\hat{y} = \arg \max(P_{\text{final}}). \quad (2)$$

The optimized α value was selected based on the accuracy of the decision-fusion network, as reported in Table 2. Consequently, α was set to 0.7 in this study, as it yielded the highest accuracy.

III. RESULTS AND DISCUSSION

Table 3 presents the evaluation metrics for the three networks compared to the existing HML network [12]. The results indicate that relying solely on video data is inadequate to recognize firefighting activity, as the mean performance scores fall below 90% in all metrics. In contrast to the existing HML sensor-based network, the Sensor Transformer network shows a notable improvement in the classification accuracy rate, with increases of 0.3% in mean precision and 1.36% in mean recall. Class imbalance is common in activity recognition tasks, so relying solely on precision or recall can be misleading. F1-score provides a comprehensive evaluation and increases only when both metrics improve together. The decision-fusion network improved performance, achieving an F1-score of 95.73%, showing significant advancement over existing methods.

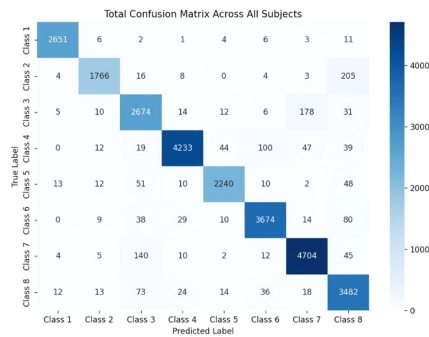


Fig. 4. Confusion matrix for the Sensor Transformer network.

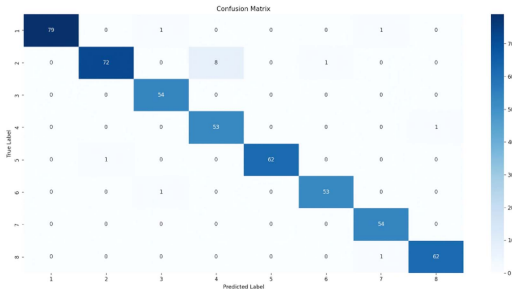


Fig. 5. Confusion matrix for the decision-fusion network.

Fig. 4 shows the confusion matrix of the Sensor Transformer network. Specifically, A3 involves walking, sometimes without upper body movement, leading to a possible misclassification as A7. Activities, such as climbing and descending floors on A4 and A6, are difficult to distinguish due to similar walking patterns in opposite directions. Interestingly, activities with similar motion patterns, such as drag movements in A2 and A5, show low misclassification rates because different objects are being dragged. More details are available in our previous work [12].

Fig. 5 illustrates the confusion matrix for the decision-fusion network, showing a mean accuracy rate of 97.02%, which exceeds that of individual networks. However, the misclassification of A2 increases, likely due to the changed decision weights, which affect the predictions after fusion. This highlights the need to adjust the fusion weights to mitigate the limitations of individual networks.

IV. CONCLUSION

This study proposes a multimodal fusion framework that utilizes DL methods to improve the classification performance of SCBA endurance training activities for firefighters based on the SFTAA dataset [12]. The results show that the Sensor Transformer performs slightly better than the existing HML network, achieving a mean F1-score of 93.31%. In addition, the Video Swin network that uses only video data exhibits the lowest performance, with all metrics below 90%. In contrast, the decision-fusion network, which combines video and sensor data, demonstrates a significant improvement in classification, achieving

96.67% mean precision, 96.13% mean recall, and 95.73% mean F1-score. This highlights the importance of multimodal fusion in advanced HAR applications, although network generalization is constrained by the focus on only SCBA endurance training activities. Future work will focus on optimizing computational efficiency for edge deployment, improving robustness in harsh scenarios, extending the network to additional firefighting training types, and exploring more advanced fusion approaches to improve generalization. Moreover, since some activities were undetectable due to camera view obstacles, future work will collect more comprehensive data to enable cross-validation of the leave-one-subject-out.

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This work involved human subjects or animals in its research. Approval of all ethical and experimental procedures and protocols was granted by The Research Integrity and Ethics Committee at the University of Nottingham Ningbo China.

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